Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

1 Introduction

Area-detector wide-angle x-ray scattering (WAXS) imaging provides a fast two-dimensional distribution of peak positions, line shapes, and relative Bragg intensities to fit the structure-factor while simultaneously collecting information about orientation, stacking, and mosaicity.[1–5] The difficulty is that the same detector features that make the measurement powerful can arise from several correlated experimental and sample-dependent causes, including finite wavelength bandwidth, beam divergence, grazing-incidence refraction, sample misalignment, mosaic spread, and disorder.[3, 6–8] One solution is a forward model that starts from a crystallographic and structural hypothesis and propogates it through the incident beam distribution, sample geometry, scattering kinematics, and detector response to predict the area-detector image for comparison with experiment. This direction of analysis keeps the assumptions that generate a feature attached to the feature itself and as the calculation is modular, individual physical quantities can be added or removed to diagnose samples.

Exemplary for this application are layered van der Waals solids and related two-dimensional materials whose interests are the crystallite orientations and phases. [4, 9, 10] In layered halides such as PbI_2, transport and device response can depend strongly on whether the layers lie parallel to the substrate.[11] In topological bismuth chalcogenides, strong c-axis alignment is needed to preserve the desired electronic response.[12] In transition-metal dichalcogenides, catalytic activity can increase when platelet edges are exposed by the film orientation.[13] Comparable orientational states also occur across a wider set of layered thin-film families, including telluride chalcogenides (e.g., Bi_2Te_3, Sb_2Te_3), dichalcogenides (e.g., MoS_2, NbS_2), layered halides and oxyhalides (e.g., BiI_3, BiOI), and laminated 2D-flake films such as $Ti_3C_2T_x$ MXenes and graphene-oxide papers.[14–21]

Illustrated in Fig.1, area-detector WAXS is attractive for this problem because a single exposure probes a large reciprocal-space volume rich with information of both structure and orientation.[1–3, 8, 22] The difficulty is that many layered films do not fall cleanly into either of the standard diffraction limits. They are neither single crystals, where reciprocal space consists of discrete points, nor fully isotropic three-dimensional (3D) powders, where orientational averaging produces spherical shells.[6] Instead, a common thin-film orientational state is the two-dimensional (2D) oriented powder, in which plate-like crystallites share an aligned layer normal while remaining azimuthally randomized in-plane.[4, 6, 23] This intermediate orientational state produces detector patterns that inherit features of both limiting cases while necessitating nontraditional analysis techniques.[6]

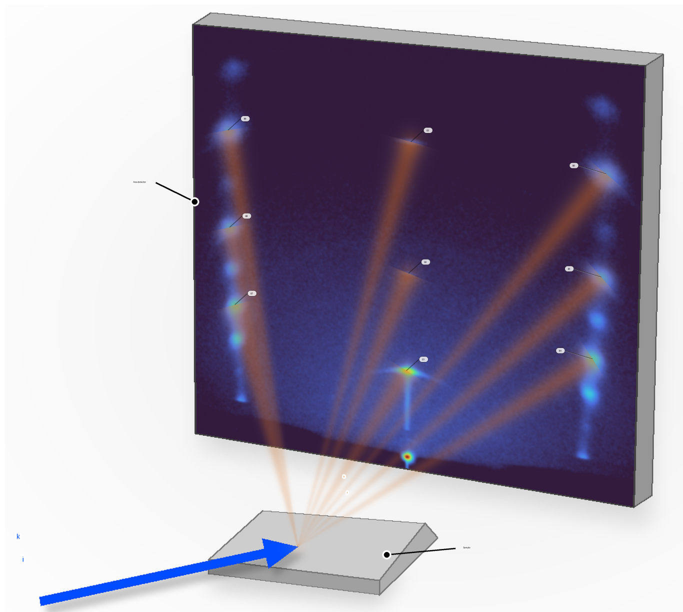

Figure 1: Illustrated of a 2D-powder WAXS measurement using a measured sample of PbI_2. The incident beam strikes the film and a diffracted intensity is recorded directly on the area detector. Bright features indicate Bragg peaks, while the streaks between them and arcs illustrate how oriented crystallites, finite mosaic spread, and detector geometry distribute reciprocal-space intensity across the detector image.

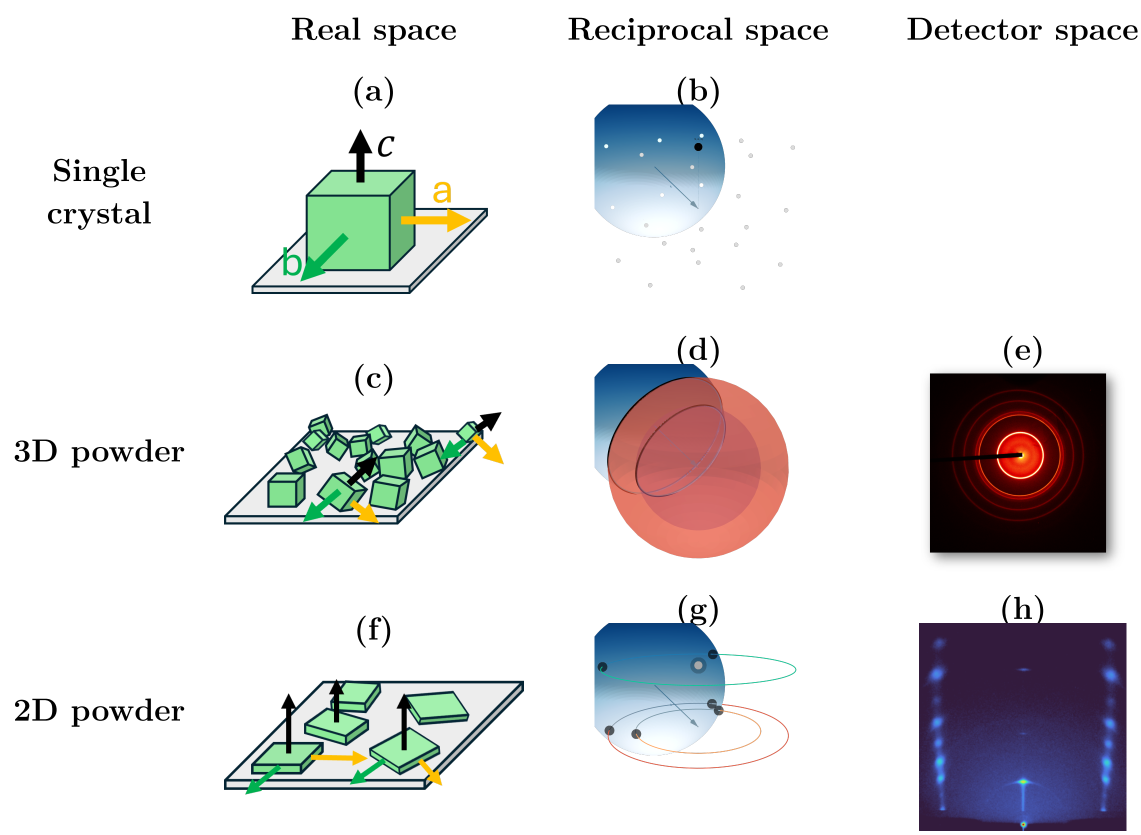

Figure 2: Diffraction across three orientational limits: single crystal, 3D powder, and 2D powder. Columns show the corresponding real-space, reciprocal-space, and detector-space views, while rows group the orientational limits. Panels (a,b) show a single crystal in real space and reciprocal space. Panels (c–e) show a 3D powder, with randomly oriented crystallites, spherical reciprocal-space shells, and Debye–Scherrer rings on the detector. Panels (f–h) show a 2D powder, with aligned surface normals and in-plane orientational disorder, reciprocal-space rings with specular Bragg features, and the corresponding detector pattern with symmetry about the central axis.

Figure 2 summarizes the real-space, reciprocal-space, and detector-space consequences of these three orientational limits. In the Ewald construction, elastic scattering requires that reciprocal-space intensity lie on the Ewald sphere of radius $|\mathbf{k}|$.[24] For a single crystal, reciprocal space consists of isolated Bragg points (Fig. 2b), so at fixed incidence angle the Ewald sphere typically intersects only a small subset of reflections and the detector records sparse spots.[24] For a 3D powder, random crystallite orientations smear the intensity of each Bragg point over a spherical shell, and the Ewald-sphere intersection produces Debye–Scherrer rings on the detector (Fig. 2d,e).[24] For a 2D powder, azimuthal disorder about the common c axis smears each off-specular Bragg point into an in-plane ring about Q_z, while specular $(00L)$ reflections remain localized near the Q_z axis (Fig. 2g).[1, 6] The corresponding illustration in Fig. 2f schematically shows 2D platelets with aligned growth, and a representative 2D-powder detector pattern is shown in Fig. 2h. When the Ewald sphere intersects this reciprocal-space distribution, the detector records hybrid signatures: pairs of symmetric spots from the off-specular ring intersections and single-crystal-like features near the specular stripe from $(00L)$ intensity.[1, 2, 25] Mosaicity then redistributes intensity, broadening the ring intersections into arcs and spreading the $(00L)$ features into cap-like intensity near the specular region.[1, 6]

This hybrid detector-space information is what makes the 2D-powder problem both rich and difficult. Many standard WAXS workflows reduce area-detector images to 1D profiles by azimuthal integration, often with masking near the direct beam and specular region.[2, 3, 26–28] Those steps suit nearly isotropic powders, but for 2D powders they deweight or remove off-specular arcs and specular caps that constrain out-of-plane correlations, stacking, and mosaic tails.[1, 2] Stacking disorder is also commonly inferred from 1D interference models and refinements,[29–31] which do not directly exploit the two-dimensional anisotropy of oriented-film scattering. More general intermediate orientation distributions can be treated with orientation-distribution-function (ODF) refinements or preferred-orientation corrections,[32–37] but those approaches are more general than needed for the narrowly constrained 2D-powder case considered here.[36, 37] Recent radial-profile approaches have made thin-film GIXD substantially more quantitative, but peak overlap and background separation can remain limiting in strongly anisotropic patterns, especially at small incidence angles.[5] For the layered films considered here, the measured morphology already motivates axial symmetry about the film normal and a low-dimensional out-of-plane mosaicity kernel.[4, 6, 23] We therefore assume this physically motivated 2D-powder orientation distribution a priori and refine it with a detector-space forward model, using off-specular arcs and specular-family line shapes as direct constraints.[1, 2]

Here we present a detector-space forward model for 2D-powder WAXS. Starting from an indexed crystallographic model, it predicts area-detector images and applies the same projection and integration procedures to measured and calculated intensity. The comparison uses detector-space peak positions, local line shapes, relative Bragg intensities, and projected Q_z profiles. Prior work has established the main ingredients needed for such a framework, including grazing-incidence scattering formalisms,[38, 39] reciprocal-space mapping and indexing for oriented thin films,[1–3, 25] and forward calculations of diffraction from textured films.[7] Complementary atomistic tools have also been developed for constructing commensurate multilayer 2D heterostructure supercells and visualizing reciprocal-space diffraction signatures, including Nookiin.[40] In contrast, the present work carries the calculation in detector space through detector geometry, grazing-incidence propagation, beam phase space, mosaicity, resolution, and ordered structure-factor weighting. The main-text validation is centered on the ordered 2D-powder limit using Bi_2Se_3 and Bi_2Te_3 films.[12, 14] After the ordered-film validation, we assemble PbI_2 as a diffuse-scattering extension in which stacking disorder changes the structure factor along fixed-Q_R rods while the same detector-space projection and mosaicity framework are retained.

2 Diffraction Model

The detector-space forward model redistributes the structure-factor intensity of the contributing reflections over mosaic distributions. By Monte Carlo sampling the incident beam profile across those orientations, weighting each ray by transmission, and mapping the summed intensity to detector pixels the calculated and measured images can then be compared using the same peak positions, line shapes, relative Bragg intensities, and projected L-integrated profiles.

2.1 Diffraction Geometry

We first consider a single incident ray striking a perfectly aligned crystallite, with no beam divergence, no refraction, and no absorption. Elastic scattering requires

$$|\mathbf{k}_i| = |\mathbf{k}_f| = \frac{2\pi}{\lambda}, \qquad \mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i.$$

Diffraction occurs when the momentum transfer matches a reciprocal-lattice vector, $\mathbf{Q} = \mathbf{G}$ as seen in Fig. 2d. In the Ewald construction, this means that the reciprocal-space point for that reflection lies on the Ewald sphere of radius $|\mathbf{k}_i|$, and the simplest diffracted ray is obtained from

$$\mathbf{k}_f = \mathbf{k}_i + \mathbf{G}.$$

For a fixed $|\mathbf{G}|$, the set of possible reflection orientations forms a Bragg sphere of radius $|\mathbf{G}|$ about the reciprocal-space origin. The allowed exit vectors are found by intersecting that Bragg sphere with the Ewald sphere and then propagating the resulting exit ray to the detector.

For each trial incident ray, the model computes the sample intersection, applies the incidence geometry and optical corrections, finds the reciprocal-space intersections that satisfy elastic scattering, and propagates the resulting exit rays to the detector. Rays are Monte Carlo sampled from the probability distribution formed by the shape, bandwidth, and divergence of the beam. From every Bragg peak's structure factor spread upon the mosaic, a second Monte Carlo sampling occurs for each ray to choose an intersection point. Footprint, detector tilt, refraction, and absorption enter as experimental resolution and intensity terms. The coordinate conventions for the lab frame, detector plane, sample rotation, and illuminated footprint are summarized in Figs. 3 and 4.

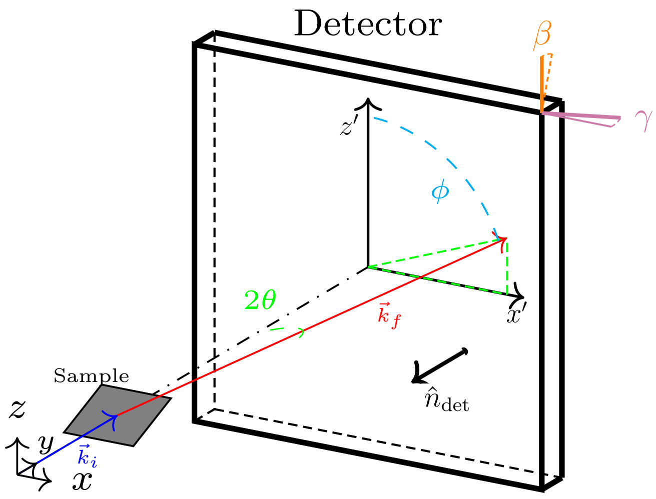

Figure 3: Overall lab-frame geometry. The incident wavevector $\mathbf{k}_i$ strikes the sample, and a representative exit wavevector $\mathbf{k}_f$ reaches the detector at scattering angle 2θ and detector-plane azimuth ϕ. The detector at distance D is tilted by β and γ.

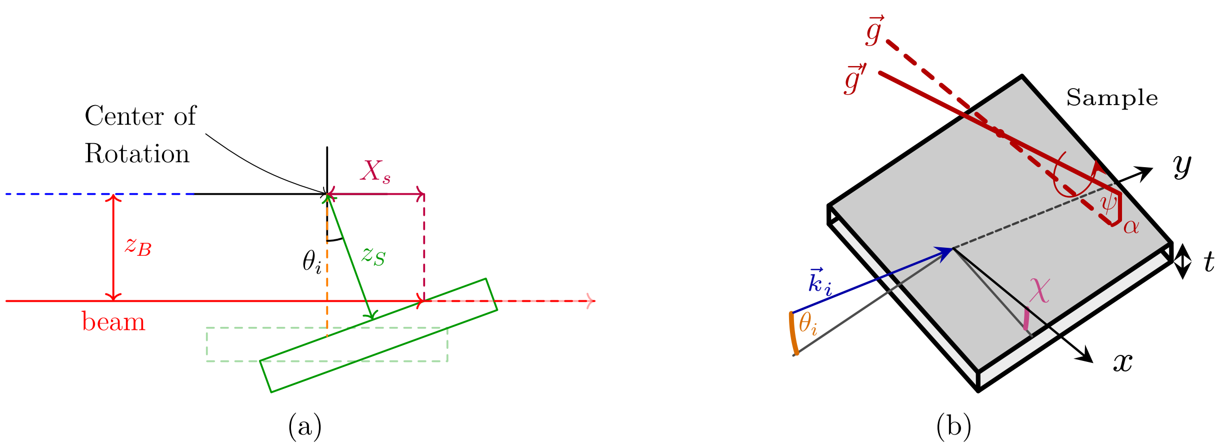

Figure 4: Two-stage sample geometry. Left: rotation about the goniometer center with sample offset z_S and beam offset z_B. θ_i is the incidence angle. Right: full alignment adds tilt δ and footprint length X_s within an illuminated window (W, H) on a film of thickness t. $\vec{g}$ is the goniometer arm (normal to $\hat{y}$), and $\vec{g}'$ is the misaligned arm obtained by rotations α and ψ.

2.2 Structure Factor

We enumerate integer in-plane Miller pairs (H, K), construct the reciprocal-space rod along G_z for each pair, and evaluate its detector intersections. For labels and profile extraction, the in-plane pair is represented by the scalar family index m, defined by the in-plane sum

$$m(H,K) = \begin{bmatrix} H & K \end{bmatrix} \mathbf{M}_{\parallel} \begin{bmatrix} H \\ K \end{bmatrix}.$$

Here $\mathbf{M}_{\parallel}$ is the in-plane reciprocal-lattice metric, with any lattice-dependent normalization absorbed into the definition of m. For the hexagonal bismuth chalcogenides, this scalar is proportional to $H^2 + H\,K + K^2$. Thus, m labels an in-plane reciprocal-space rod family. In 2D-powder calculations, all integer (H,K) rods with the same m, and therefore the same Q_r, are evaluated individually and then summed after projection. Multiplicity and orientation effects therefore emerge within the forward model.

For each reflection with reciprocal-lattice vector

$$\mathbf{G} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*,$$

we compute the structure factor as

$$F_{\mathbf{G}} = \sum_a o_a\, f_a(Q)\, \exp[2\pi i\, \mathbf{G}\cdot\mathbf{r}_a]\, \exp\Big[-2\pi^2\, \mathbf{G}\cdot\mathbf{U}_a\cdot\mathbf{G}\Big]\,, \tag{1}$$

where the sum runs over atoms a in the unit cell. Here o_a, $f_a(Q)$, $\mathbf{r}_a$, and $\mathbf{U}_a$ are the occupancy, atomic form factor, fractional position, and anisotropic displacement tensor for atom a. Symmetry constraints are retained during refinement, so only the independent crystallographic parameters are varied. The detector weight for each allowed ray is multiplied by $|F_{\mathbf{G}}|^2$ before mosaic and experimental broadening are applied.

2.3 Mosaicity

Mosaicity is the out-of-plane orientational spread of the layered crystallites about the mean film normal illustrated in Fig. 2(a,b,c). It is distinct from the random in-plane rotation that defines the two-dimensional powder limit. Random in-plane rotation generates the equivalent m-indexed rod family at a common in-plane reciprocal-space radius Q_R, while mosaicity determines which tilted members of that family can satisfy the Ewald condition at a fixed incident angle.

Shown in Fig. 5 is a full two-dimensional area-detector image of Bi_2Se_3 at $\theta_i = 5°$ with the low-L, $m = 0$ specular region expanded. A single incident angle should select only a limited part of the specular $m = 0$ family if the film has only a narrow orientational distribution. The measured image instead contains unexpected $00L$ intensity at $L = 3$ and $L = 6$, nominally near 6° and 12° in 2θ, respectively. Their spread in ϕ is direct evidence for crystallites tilted away from the mean layer normal, and their visibility at this incidence condition requires a long mosaic tail in addition to the narrow mosaic core. The near-origin $L \approx 0$ intensity is due to reflectivity and is retained in the crop to clarify how it can obscure the nearby 003 and 006 features used to interpret mosaicity and bandwidth; its treatment is given in Sec. 2.5. The 2θ breadth of the $L = 3$ feature is not assigned to mosaicity alone. Sec. 2.4 explains how finite wavelength bandwidth broadens the Ewald condition for small Bragg spheres.

Detector Image features of Bi_2Se_3

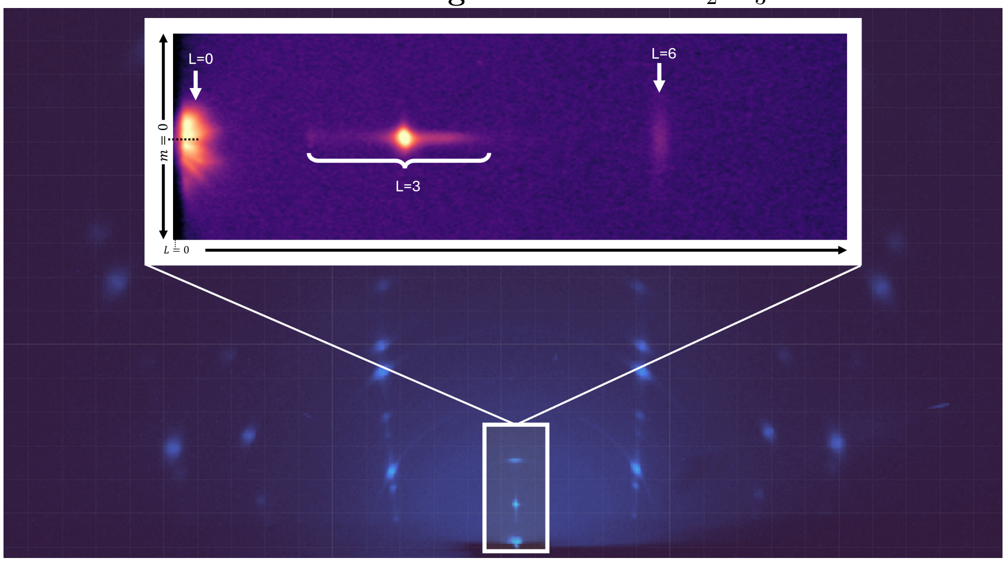

Figure 5: Close-up of the low-L $m = 0$ detector feature from the 5° Bi_2Se_3 image. The bright near-origin intensity is assigned to near-critical reflectivity and is treated separately in Sec. 2.5. The star-like morphology of the $L = 3$ feature and the appearance of the $L = 6$ peak are not expected from an ideal single-crystallite calculation or from only a narrow mosaic distribution. These higher-L features reflect the combined effect of wavelength bandwidth and angular divergence, which give the Ewald sphere an effective thickness, and a long mosaic tail, which supplies tilted crystallites that bring otherwise off-condition specular-family intensity into the measured detector region.

The in-plane rod-family arcs are described by a Gaussian mosaic distribution. The specular-family features expose a different part of the orientational distribution: weak intensity from crystallites tilted far enough from the mean normal to satisfy reflections that would otherwise be missed at the nominal incidence condition. A rapidly decaying Gaussian-like mosaic density can reproduce the central widths of near-condition features but does not provide enough tilted-crystallite population in the tails. We therefore use a compact two-component angular density,

$$\omega_{\mathrm{w}}(\Delta\theta;\eta,\Gamma,\sigma) = \eta L_{\mathrm{w}}(\Delta\theta;\Gamma) + (1-\eta)G_{\mathrm{w}}(\Delta\theta;\sigma), \qquad 0 \le \eta \le 1. \tag{2}$$

Here G_{w} is the Gaussian-like core, L_{w} is the Lorentzian-like tail, η is the Lorentzian weight, and $\Delta\theta$ is the crystallite tilt relative to the mean layer normal. The two widths are independent with shared centers.

Figure 6 summarizes how the orientational distribution manifests for $m = 0$ and $m \neq 0$ in reciprocal space. For $m \neq 0$, random in-plane rotation and out-of-plane mosaicity produce off-specular arcs. For $m = 0$, the same angular distribution produces cap-like $00L$ intensity near the specular direction, where the tail contribution is especially visible.

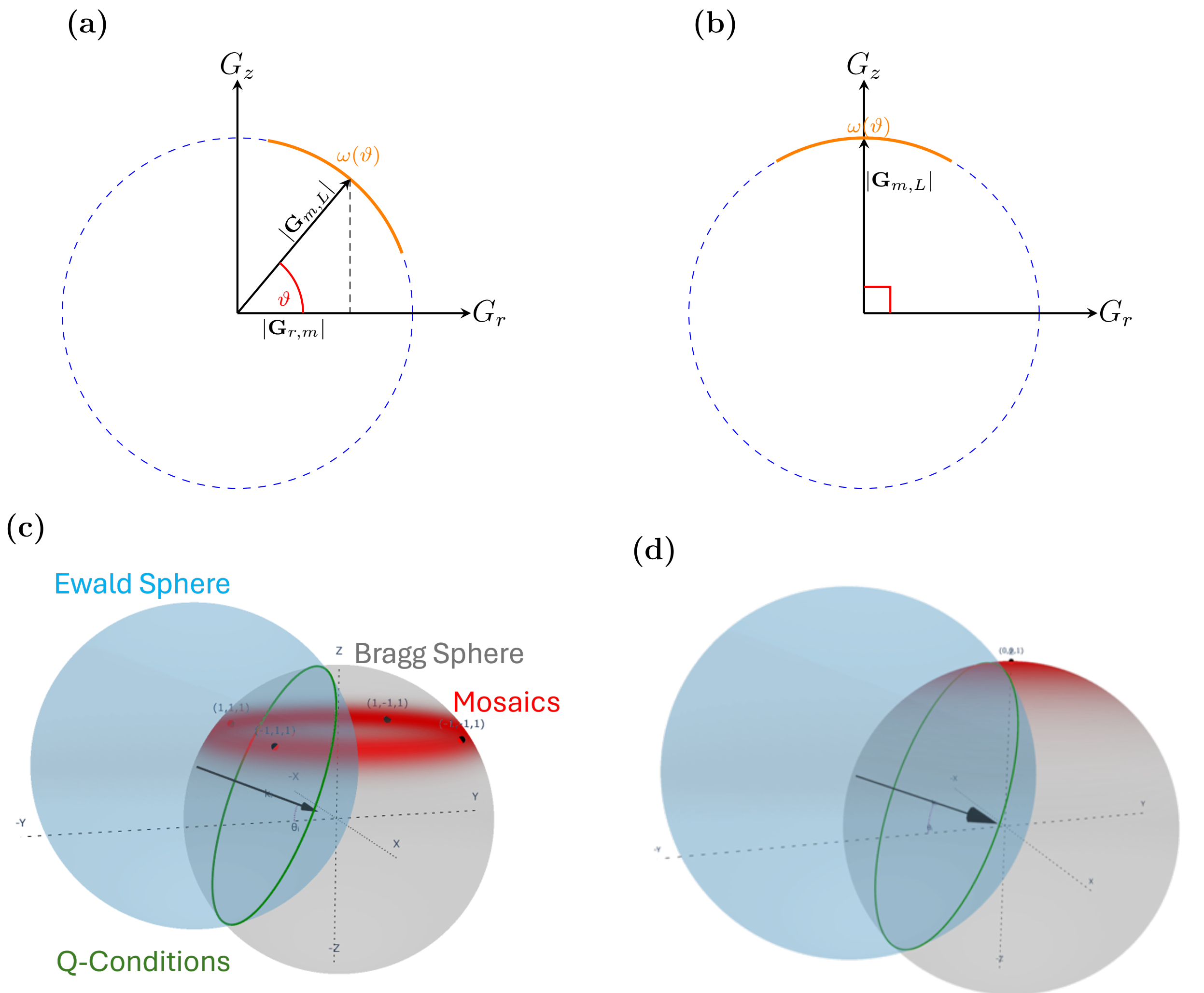

Figure 6: Mosaic broadening in reciprocal space and detector space. (a,c) For $m \neq 0$, random in-plane rotation and finite out-of-plane mosaicity give off-specular arcs or bands. (b,d) For $m = 0$, the same orientational distribution gives cap-like $00L$ intensity near the specular direction. The narrow Gaussian-like core controls the main width of near-condition features, while the Lorentzian-like tail supplies the weak tilted-crystallite population needed for off-condition $m = 0$ peaks.

2.4 Bandwidth-Related Effects

The anomalous intensity along 2θ for the $m = 0$, $L = 3$ feature in Fig. 5 can be explained by the bandwidth forming an Ewald shell on a much smaller Bragg sphere. In this regime the measured intensity is not set by an external Ewald-sphere intersection alone.

For each sampled wavelength λ_j, with spectral weight w_j, the model evaluates

$$k_0(\lambda_j) = \frac{2\pi}{\lambda_j}, \qquad \tilde{n}(\lambda_j) = 1 - \delta_n(\lambda_j) + i\beta_n(\lambda_j). \tag{3}$$

Here δ_n is the refractive decrement that shifts the internal phase velocity, and β_n is the absorptive correction that gives the finite penetration depth. The real decrement may be estimated from the electron density ρ_e and classical electron radius r_e, while the imaginary term is the wavelength-dependent optical input used directly by the complex wavevector calculation:

$$\delta_n(\lambda_j) = \frac{r_e \lambda_j^2 \rho_e}{2\pi}, \qquad \beta_n(\lambda_j) = \mathrm{Im}\{\tilde{n}(\lambda_j)\}. \tag{4}$$

Thus bandwidth changes both the Ewald-sphere radius and the optical constants that determine refraction and attenuation. For an incident wavevector in air, the grazing angle and azimuth are

$$\sin\theta_i = \frac{k_{i,z}}{k_0(\lambda_j)}, \qquad \tan\varphi_i = \frac{k_{i,x}}{k_{i,y}}. \tag{5}$$

The transmitted angle and internal wavevector scale are

$$\theta_t(\lambda_j) = \cos^{-1}\left[\frac{\cos\theta_i}{\mathrm{Re}\{\tilde{n}(\lambda_j)\}}\right], \tag{6}$$

$$k'(\lambda_j) = k_0(\lambda_j)\left[\mathrm{Re}\{\tilde{n}(\lambda_j)^2\}\right]^{1/2}. \tag{7}$$

The in-plane internal components are fixed by θ_t and φ_i. The remaining surface-normal component is complex. Using the closed-form solution of the $a + ib$ decomposition, define

$$a_t(\lambda_j) = k'(\lambda_j)^2 \sin^2\theta_t(\lambda_j), \tag{8}$$

$$b(\lambda_j) = k_0(\lambda_j)^2\, \mathrm{Im}\{\tilde{n}(\lambda_j)^2\}. \tag{9}$$

Then

$$\mathrm{Re}\{k_{t,z}(\lambda_j)\} = \left[\frac{\sqrt{a_t(\lambda_j)^2 + b(\lambda_j)^2} + a_t(\lambda_j)}{2}\right]^{1/2}, \tag{10}$$

$$\mathrm{Im}\{k_{t,z}(\lambda_j)\} = \left[\frac{\sqrt{a_t(\lambda_j)^2 + b(\lambda_j)^2} - a_t(\lambda_j)}{2}\right]^{1/2}. \tag{11}$$

The branch is chosen so that $\mathrm{Im}\{k_{t,z}\} \geq 0$, giving a decaying wave inside the film. The same result is applied to the outgoing internal wavevector using the scattered internal angle. Optical reversibility then refracts the scattered ray back into air. The algebraic derivation and the interface transmission factors are given in the Supporting Information.

The imaginary normal components set the propagation loss. For each accepted candidate, define

$$\kappa_i(\lambda_j) = \mathrm{Im}\{k_{t,z}(\lambda_j)\}, \qquad \kappa_f(\lambda_j) = \mathrm{Im}\{k'_{t,z}(\lambda_j)\}. \tag{12}$$

If L_{in} and L_{out} are the entry and exit propagation lengths used for that candidate, the absorption/evanescence weight is

$$A_{\mathrm{prop}}(\lambda_j) = \exp[-2\kappa_i(\lambda_j)L_{\mathrm{in}}(\lambda_j) - 2\kappa_f(\lambda_j)L_{\mathrm{out}}(\lambda_j)]\,. \tag{13}$$

This is the form used by the simulation: the path lengths set how far the wave propagates in the film, and the imaginary parts of the internal normal wavevectors set the damping per unit length. For a uniformly scattering film of thickness t, this may be written as a depth average,

$$\bar{A}_{\mathrm{prop}}(\lambda_j) = \frac{1}{t}\int_0^t \exp[-2\,(\kappa_i(\lambda_j) + \kappa_f(\lambda_j))\,\zeta]\,d\zeta = \frac{1 - \exp[-2\,(\kappa_i(\lambda_j) + \kappa_f(\lambda_j))\,t]}{2\,(\kappa_i(\lambda_j) + \kappa_f(\lambda_j))\,t}. \tag{14}$$

This attenuation is multiplied by the entrance and exit transmission factors and the structure-factor intensity.

The detector image is therefore an incoherent sum over wavelength samples,

$$I_{\mathrm{det}}(x, y) = \sum_j w_j\, I_{\mathrm{det}}[x, y; \lambda_j, \tilde{n}(\lambda_j)]\,. \tag{15}$$

The bandwidth has two roles: it thickens the reciprocal-space intersection through $k_0(\lambda_j)$, and it changes the refracted wave through $\tilde{n}(\lambda_j)$.

For the $00L$ family, $G_{00L} = 2\pi L/c$. A fixed wavelength spread gives an effective Ewald-shell thickness, and its angular overlap with the mosaic-populated Bragg cap scales approximately as $1/L$. Low-L Bragg spheres therefore retain overlap with the mosaic cap over a broader range of incident-angle detuning than high-L Bragg spheres. The long mosaic tail supplies the tilted crystallites, while wavelength-dependent refraction and attenuation determine how much of that intensity reaches the detector.

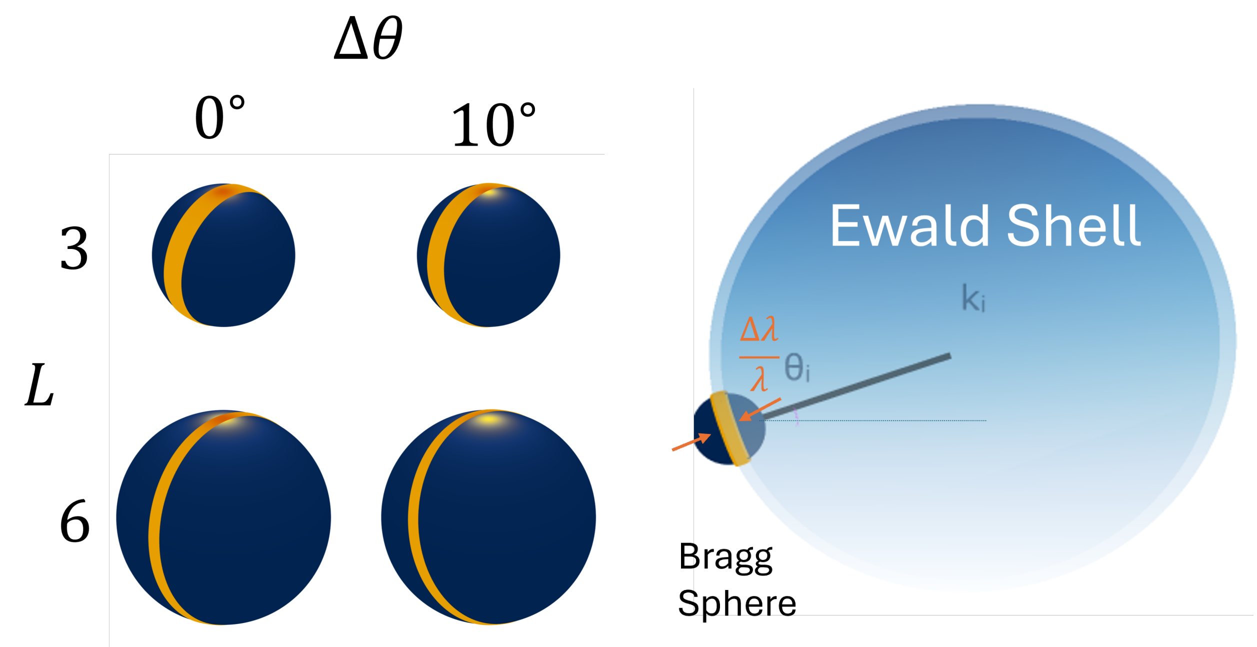

Figure 7: Bandwidth-related effects for the low-L $m = 0$ feature. The wavelength distribution changes both the Ewald-sphere radius and the refractive index used to calculate the internal wavevector. Because $G_{00L} \propto L$, a fixed bandwidth samples a larger angular fraction of low-L Bragg spheres than high-L Bragg spheres.

2.5 Reflectivity

The bright near-origin $m = 0$ intensity in Fig. 5 lies in the near-critical specular region.

The specular-family profiles in this near-critical region are calculated with an optical-to-kinematic handoff. The low-Q_z branch is calculated with a Parratt multilayer reflectivity recursion, which retains refraction, evanescence, and multiple reflection at the film interfaces.[41] At larger Q_z/Q_c, where critical-angle corrections become small, this optical branch is matched to a normalized finite-stack kinematic term. In the calculation, the two limits are joined with a smooth log-intensity blend over the region where they agree. Figure 8 overlays the measured near-critical $m = 0$ profile with the Parratt and kinematic limiting calculations on the same axes, while omitting the internal blended curve. The direct overlay shows which limit describes each part of the measured profile and where the two calculated branches converge. This construction prevents near-critical specular intensity from being treated as ordinary high-Q_z kinematic scattering while preserving the finite-stack modulation that controls the higher-L rod profile. Details of the recursion, scale estimate, and blend criterion are provided in the Supporting Information.

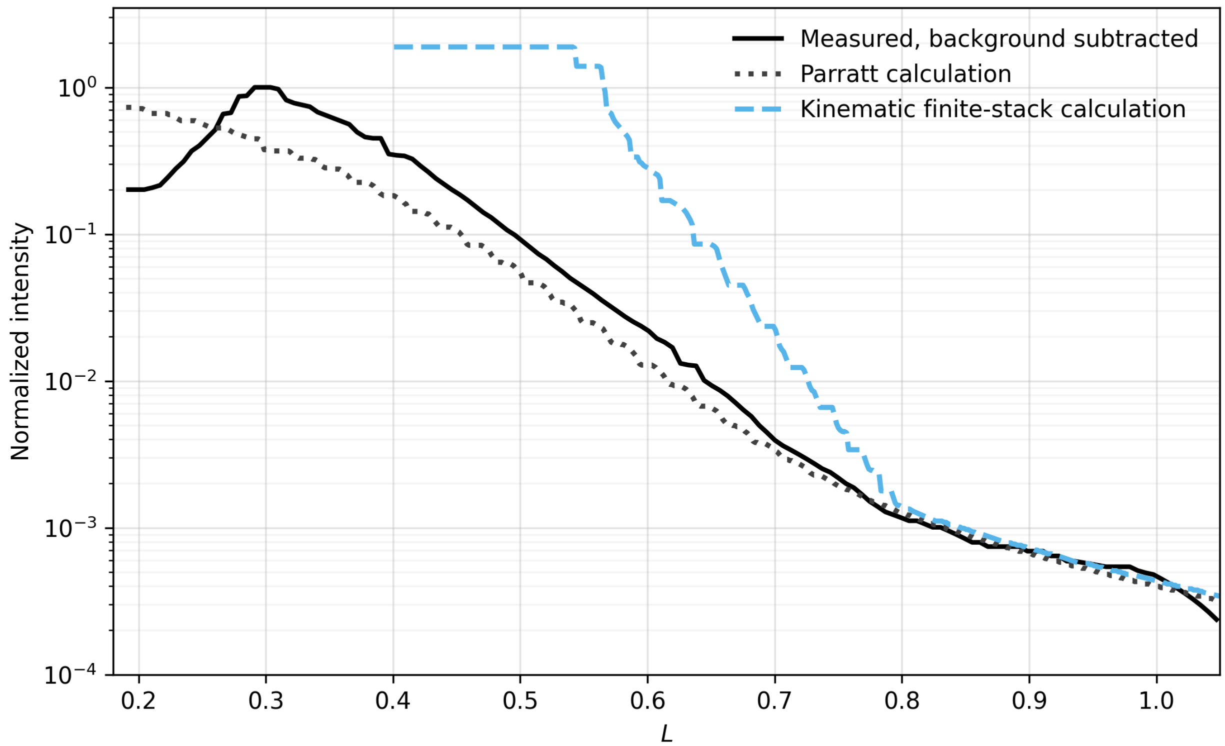

Figure 8: Measured near-critical Bi_2Se_3 $m = 0$ profile overlaid with the two limiting calculations used by the optical-to-kinematic handoff. The solid black curve is the background-subtracted detector-derived data, the dotted curve is the Parratt reflectivity calculation, and the dashed curve is the finite-stack kinematic calculation. All three are shown on the same normalized-intensity axis over $0.2 \leq L \leq 1.05$, before the 003 peak. The common overlay makes the low-L optical regime and the convergence of the calculated branches toward the measured high-L tail directly visible. The smoothly blended curve used internally by the model is omitted.

3 Results: ordered Bi_2Se_3 and Bi_2Te_3 films

Ordered Bi_2Se_3 and Bi_2Te_3 films grown by TEM were used to validate the model by showing that the same detector-space forward model quantitatively reproduces measured line shapes and intensities. Three detector images, collected at incident angles of 5°, 10°, and 15°, were used in the refinement. The 5° condition gives the clearest single-view display of both the off-specular arc families and the specular $m = 0$ features. Together, the data test the two-dimensional powder orientational limit, sample geometry, finite beam phase space, mosaicity, detector resolution, and ordered structure-factor weighting.

The detector-space panels and profile panels in Fig. 9 use the same extraction convention. In panels (a) and (b), the solid colored curves are the calculated centerlines of selected m-indexed Bragg-rod trajectories on the detector. The semi-transparent colored bands are the finite detector regions integrated to form the one-dimensional profiles. The dashed colored outlines mark the edges of those integration bands. The central blue band is the $m = 0$ specular-family trajectory, while the paired off-specular bands are the two detector-side intersections of the same two-dimensional powder rod family. The plus and minus labels identify those two detector-side branches only.

Panels (c) and (d) show the corresponding line profiles after applying the same projection and integration operation to the measured and calculated images. The horizontal coordinate is L, so a marker labeled $L = n$ denotes the nominal $00n$ or corresponding off-specular Bragg condition for that selected rod family. The black solid curves are the measured profiles. The orange dashed curves are the simulated profiles after the same detector-space projection. The off-specular rows are plotted on a linear intensity scale, while the $m = 0$ row is plotted on a logarithmic scale because the important result is the persistence of weak, unexpected $00L$ intensity far from the strongest specular peak.

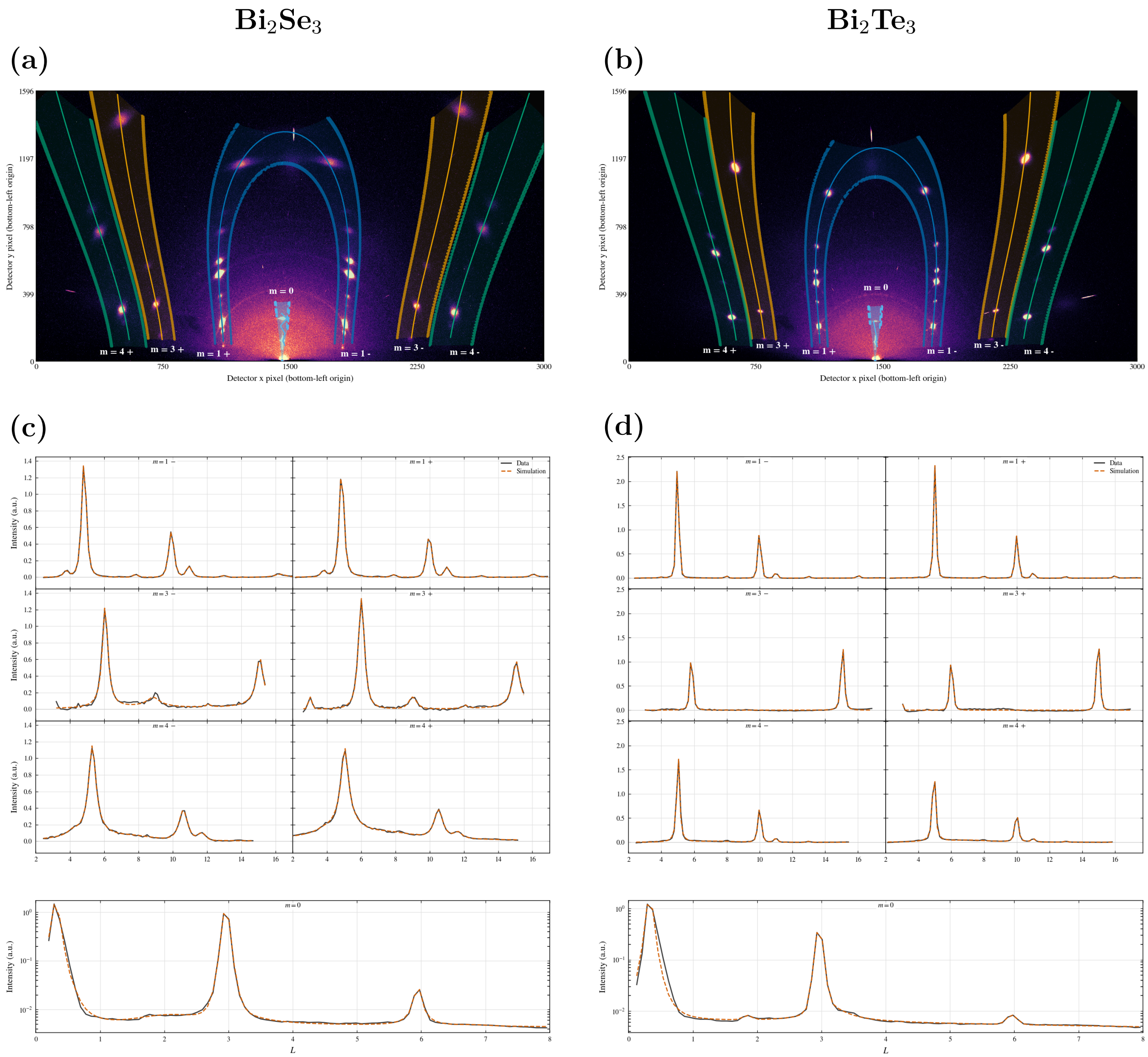

Figure 9: Ordered-film validation of Bi_2Se_3 and Bi_2Te_3 at $\theta_i = 5°$. (a,b) Measured detector images with selected reciprocal-space extraction regions overlaid. Solid colored curves are the trajectory centerlines. Semi-transparent colored regions are the detector pixels integrated to form the profiles. Dashed colored outlines mark the integration-band boundaries. The labels m identify two-dimensional powder rod families, and the signs label the two detector-side branches of the same family. The bright low-L intensity near the critical-angle region is assigned to reflectivity caught by finite bandwidth. (c,d) Extracted profiles plotted versus L after applying the same projection to measured and calculated images. Black solid curves are measured data and orange dashed curves are simulations. The off-specular rows test detector geometry, the two-dimensional powder construction, the mosaic-core width, and the ordered structure-factor pattern. The semilogarithmic $m = 0$ rows emphasize the unexpected persistence of additional $00L$ peaks, which requires a long Lorentzian mosaic tail in addition to the narrow mosaic core.

Because mosaicity, beam divergence, wavelength spread, incidence-angle uncertainty, detector resolution, and finite integration width all broaden the effective scattering condition, the projected coordinate should not be interpreted as an ideal reciprocal-space cut from a single perfect crystallite. The comparison remains meaningful because the calculated image is subjected to the same projection and resolution effects before it is compared with the measured profile. The figure therefore tests the detector-space calculation as used experimentally. This distinction is especially important for the unexpected $00L$ peaks and the enhanced low-L specular intensity, where incidence angle, wavelength bandwidth, divergence, detector-coordinate uncertainty, background, and the Lorentzian mosaic tail are most strongly coupled.

In Fig. 9, the off-specular and $m = 0$ profiles play complementary roles. The off-specular arcs check the two-dimensional powder reciprocal-space construction and calibrated detector geometry, while their widths and asymmetries constrain the Gaussian-like mosaic core and instrumental resolution. The off-condition $m = 0$ peaks test the Lorentzian mosaic tail and show why a rapidly decaying mosaic core alone is insufficient. The ordered-film result is therefore not only that the fitted curves match the strongest peaks, but that the same physical model also accounts for weaker and less intuitive detector features that were not built into a one-dimensional reduction.

4 Results: diffuse scattering and stacking disorder in PbI_2

Figure 10 shows a CVD-grown PbI_2 film and a crop spanning the $m = 0$ and $m = 1$ Bragg rods. The $m = 1$ rod contains measurable intensity between the integer Bragg maxima, whereas the $m = 0$ rod is dominated by sharp peaks and finite-thickness oscillations. This rod-dependent redistribution of intensity is the experimental motivation for the stacking-disorder model. In a two-dimensional powder, random in-plane rotation maps each in-plane reflection family onto a reciprocal-space cylinder at fixed Q_R. Perfect stacking concentrates intensity at discrete Q_z values on a cylinder; loss of stacking coherence redistributes part of that intensity along the same cylinder.

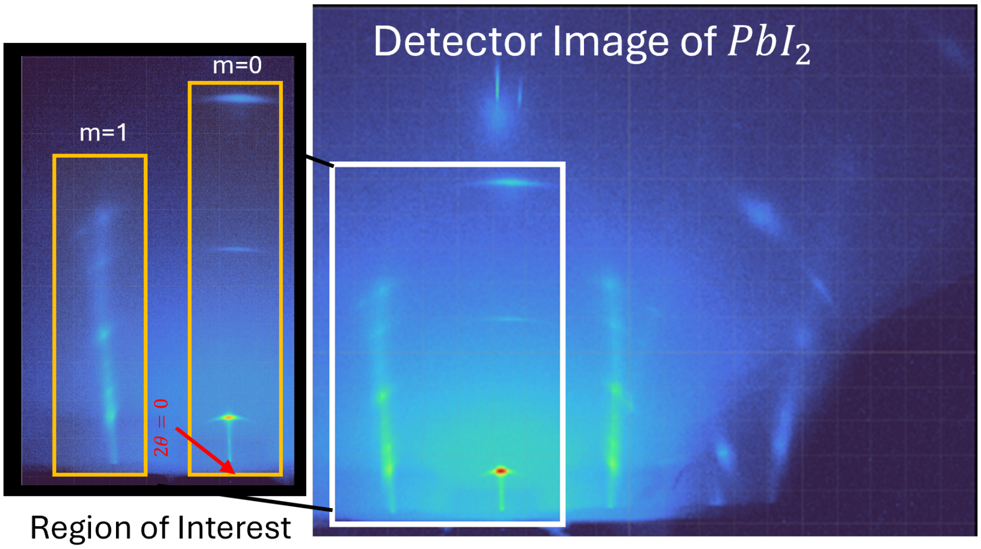

Figure 10: Measured PbI_2 diffraction image (right) and an enlarged region of interest (left). The direct-beam position defines the vertical $m = 0$ trajectory. Diffuse intensity is visible between the ordered maxima on the neighboring $m = 1$ rod, while the $m = 0$ profile remains insensitive to the lateral registry shifts that generate stacking contrast on other rods.

4.1 Direction-resolved transition-matrix model

The structural unit is one I–Pb–I trilayer. Its state is written $\alpha = (r, \sigma)$, where $r \in \{0, 1, 2\}$ denotes one of three basal registries and $\sigma \in \{+, -\}$ denotes the two trilayer orientations. The six states are therefore

$$\mathcal{S} = [0F_+, 1F_+, 2F_+, 0F_-, 1F_-, 2F_-] \,. \tag{16}$$

For a fixed m rod, a forward registry step $\mathbf{s} = (1/3, 2/3)$ contributes

$$\omega_{hk} = \exp[2\pi i(hs_x + ks_y)] = \exp\left[\frac{2\pi i}{3}(h + 2k)\right], \tag{17}$$

while the two local orientations carry form factors $F_+(h, k, L)$ and $F_-(h, k, L)$.

Adjacent trilayers obey the first-order interface law

$$\boldsymbol{\theta} = (a, b_+, b_-, d_+, d_-), \qquad a + b_+ + b_- + d_+ + d_- = 1. \tag{18}$$

Here a denotes no registry or orientation change. b_+ and b_- are forward and backward registry shifts without an orientation change. d_+ and d_- are the two handed shift-plus-orientation-change events. A change of orientation without a registry shift is excluded. The selected deterministic limits are $a = 1$ for 2H, $d_+ = 1$ for 4H, and $b_+ = 1$ for 6H. The opposite registry directions generate their symmetry-related handed counterparts. Figure 11 renders these cycles as projected I–Pb–I stacks. The sequences are read from bottom to top: $0F_+ \to 0F_+$ for 2H, $0F_+ \to 1F_- \to 0F_+$ for 4H, and $0F_+ \to 1F_+ \to 2F_+ \to 0F_+$ for 6H. These parent structures and PbI_2 polytypism are established experimentally.[42, 43]

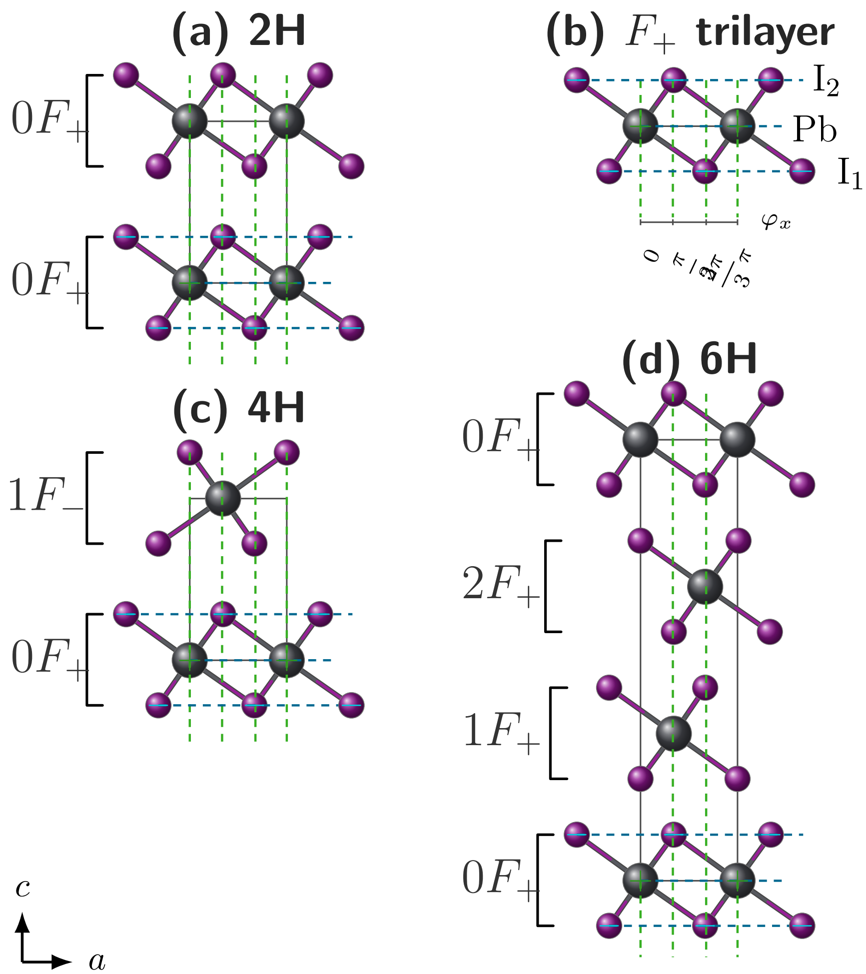

Figure 11: Projected I–Pb–I trilayers for the deterministic PbI_2 parent cycles used in the transition model. (a) 2H repeats $0F_+$ under $a = 1$. (b) An isolated F_+ trilayer shows the Pb and I atomic planes used to define the local orientation. (c) 4H alternates $0F_+$ and $1F_-$ under $d_+ = 1$. (d) 6H advances $0F_+ \to 1F_+ \to 2F_+ \to 0F_+$ under $b_+ = 1$. The c axis points upward, so each stack is read from bottom to top. In a state label rF_σ, $r \in \{0, 1, 2\}$ is the basal registry and $\sigma \in \{+, -\}$ is the trilayer orientation. Violet and gray spheres denote I and Pb, green vertical guides mark lateral registry, blue horizontal guides mark atomic planes, and brackets group each I–Pb–I trilayer. Signs are shown on the reference F_+ trilayer.

The general layer-correlation treatment follows the established transition-matrix theory of one-dimensionally disordered crystals, whereas the six states and allowed PbI_2 interface events above are the material-specific construction used here.[29, 44–46] Because the registry labels enter a given rod only through 1, ω_{hk}, and ω_{hk}^{-1}, the full six-state matrix reduces exactly to the following phase-weighted matrix in the orientation basis:

$$M(\omega) = \begin{pmatrix} a + b_+\omega + b_-\omega^{-1} & d_+\omega + d_-\omega^{-1} \\ d_+\omega^{-1} + d_-\omega & a + b_+\omega + b_-\omega^{-1} \end{pmatrix}. \tag{19}$$

The probability matrix $P = M(1)$ propagates the orientation populations, while $M(\omega)^\Delta$ sums the probabilities and registry phases of all paths connecting two trilayers separated by Δ interfaces. The complete construction and its reduction are given in Supplementary Note S3..

For each parent-rich population $j \in \{2H, 4H, 6H\}$, the disorder parameter has a direct probabilistic meaning:

$$p_j(\text{parent event}) = 1 - \epsilon_j, \qquad p_j(\text{each of the four alternative events}) = \frac{\epsilon_j}{4}. \tag{20}$$

For 2H, 4H, and 6H the parent event is a, d_+, and b_+, respectively. Written in the order (a, b_+, b_-, d_+, d_-), the five probabilities in Eq. (20) form $\boldsymbol{\theta}_j$ and are inserted directly into $M_j(\omega)$ and $P_j = M_j(1)$. Thus ϵ_j changes the matrix powers that determine how much phase coherence remains between increasingly distant trilayers.

For one definite sequence $s_n = (r_n, \sigma_n)$, the layer amplitude is $g_{s_n} = \omega^{r_n} F_{\sigma_n}$ and the coherent stack amplitude is $A_N(L) = \sum_{n=0}^{N-1} \xi(L)^n g_{s_n}$, where $\xi(L) = \exp(iq_z d)$ is the vertical phase between neighboring trilayer centers. Expanding $\langle |A_N|^2 \rangle$ leaves two familiar contributions: the intensity scattered by each trilayer individually and the interference between distinct trilayer pairs.

For population j, define the orientation population of trilayer m as

$$\boldsymbol{\pi}_{j,m} = \left(p^+_{j,m}, p^-_{j,m}\right) = \boldsymbol{\pi}_{j,0} P_j^m, \qquad p^+_{j,m} + p^-_{j,m} = 1. \tag{21}$$

Here j identifies the 2H-, 4H-, or 6H-rich population and m identifies a trilayer within the finite stack. The quantities $p^+_{j,m}$ and $p^-_{j,m}$ are ordinary probabilities that trilayer m has orientation F_+ or F_-, respectively; they are neither scattering amplitudes nor the sample weights w_j used below.

The disorder-averaged interference amplitude for trilayers m and $m + \Delta$ is

$$K^{(j)}_{m,\Delta} = \begin{pmatrix} p^+_{j,m} & p^-_{j,m} \end{pmatrix} \begin{pmatrix} F^*_+ & 0 \\ 0 & F^*_- \end{pmatrix} M_j(\omega)^\Delta \begin{pmatrix} F_+ \\ F_- \end{pmatrix}. \tag{22}$$

The finite-stack intensity is then written as "self scattering plus pair interference":

$$I_j(L) = I_{j,\mathrm{self}}(L) + I_{j,\mathrm{pair}}(L), \tag{23a}$$

$$I_{j,\mathrm{self}}(L) = \frac{1}{N}\sum_{m=0}^{N-1}\left[p^{+}_{j,m}|F_+|^2 + p^{-}_{j,m}|F_-|^2\right], \tag{23b}$$

$$I_{j,\mathrm{pair}}(L) = \frac{2}{N}\,\mathrm{Re}\sum_{\Delta=1}^{N-1}\xi(L)^{\Delta}\sum_{m=0}^{N-1-\Delta}K^{(j)}_{m,\Delta}. \tag{23c}$$

The self term averages the isolated-trilayer intensity over the two possible orientations. The pair term groups all trilayer pairs by their separation Δ: there are $N-\Delta$ pairs at that separation, and each carries the same vertical phase $\xi(L)^{\Delta}$. In Eq. (22), the expression can be read from right to left. The final column supplies the scattering amplitude of the later trilayer, $M_j(\omega)^{\Delta}$ sums all allowed stacking paths and registry phases between the two layers, the diagonal matrix supplies the conjugated amplitude of the earlier trilayer, and $\boldsymbol{\pi}_{j,m}$ averages over whether that earlier trilayer is in the F_+ or F_- orientation. The factor $2\,\mathrm{Re}$ in Eq. (23c) includes the reverse-ordered conjugate pair without writing it separately. The complete derivation is given in Supplementary Note S3.

The 2H-, 4H-, and 6H-rich domains are taken to be mutually incoherent, so the three calculated intensities (not their amplitudes) are combined as

$$I_{\mathrm{calc}}(L) = S\left\{\left[w_{2H}I_{2H}(L) + w_{4H}I_{4H}(L) + w_{6H}I_{6H}(L)\right]\otimes R(L)\right\}, \qquad w_{2H}+w_{4H}+w_{6H}=1. \tag{24}$$

Each $I_j(L)$ is first evaluated from Eq. (23) using its own ϵ_j. The weights then specify the fraction of the total calculated intensity assigned to each parent-rich population, the shared resolution function $R(L)$ broadens the combined profile, and S converts the normalized calculation to the measured intensity scale. Accordingly, ϵ_j is an effective per-interface disorder probability under the stated equal-alternative template, whereas w_j is an effective population weight.

For $h=k=0$, $\omega_{hk}=1$ and $F_+=F_-$. Registry shifts and orientation changes then leave the scattering amplitude unchanged, so stacking faults are contrast-matched rather than absent. The oscillations on the $m=0$ rod are therefore finite-thickness fringes. The same criterion applies to any other rod for which both the registry phase and the orientation form factors are indistinguishable.

4.2 Refinement

The disorder stage is applied only after the detector geometry, sample alignment, mosaic distribution, lattice constants, ordered PbI_2 structure factor, and profile resolution have been fixed by the upstream refinement. Measured and calculated detector images are projected through the same constant-Q_R rod masks and reported versus Q_z or the corresponding stacking coordinate L. No independent peak areas are introduced before the stacking model is evaluated, because doing so would absorb the diffuse between-peak intensity that the transition law is intended to explain.

The core disorder parameters are

$$\boldsymbol{\Theta}_{\mathrm{dis}} = (\epsilon_{2H}, \epsilon_{4H}, \epsilon_{6H}, w_{2H}, w_{4H}, w_{6H}, N), \qquad w_{2H} + w_{4H} + w_{6H} = 1, \tag{25}$$

so the three weights contain only two independent degrees of freedom. A stable refinement can first impose a shared disorder magnitude $\epsilon_{2H} = \epsilon_{4H} = \epsilon_{6H}$ and release the three values only when the retained fault-sensitive rods support phase-dependent broadening. The selected-profile residual is

$$r_j = \frac{I_j^{\mathrm{meas}} - I_j^{\mathrm{model}}(\boldsymbol{\Theta}_{\mathrm{dis}})}{\sigma_j}, \tag{26}$$

with any allowed global scale or low-order background terms declared separately. The fit is accepted only when it improves the selected diffuse profiles without degrading the ordered detector-space agreement established in the preceding stages.

Disorder Series Fitting

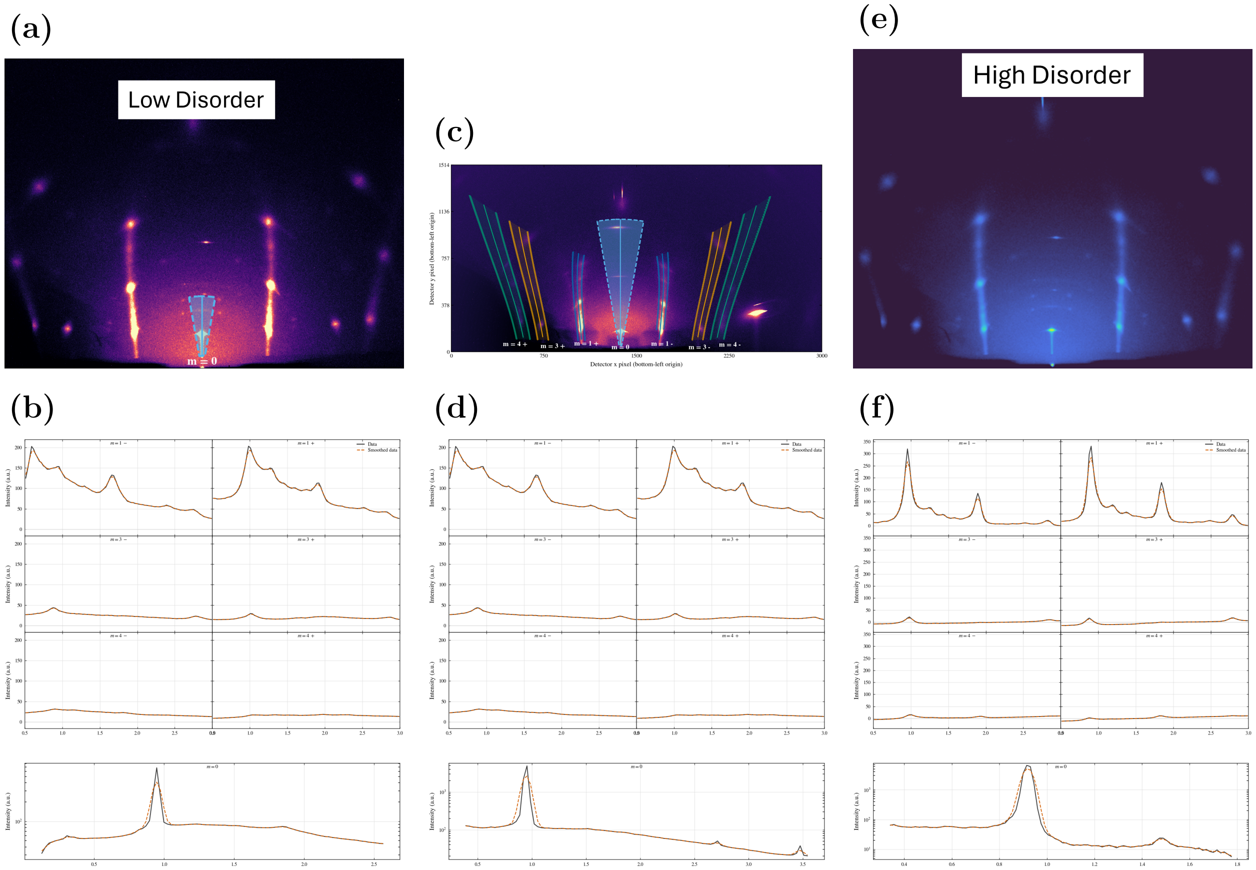

Figure 12: PbI_2 diffuse-rod comparison at $\theta_i = 4°$ for samples spanning increasing stacking disorder. The top row shows measured detector images and the constant-Q_R extraction regions. Solid curves mark the projected rod centerlines, shaded bands mark the integrated detector pixels, and dashed outlines mark the integration boundaries. The bottom row compares profiles obtained by applying the same projection to data and calculation; black solid curves are measured and orange dashed curves are calculated. The series tests whether one detector-space geometry and ordered baseline, augmented by the transition-matrix stacking model, reproduce the progressive transfer of intensity from ordered maxima into the diffuse rod signal.

Figure 12 shows the visible progression from sharp ordered maxima to stronger between-peak intensity. The absence of comparable diffuse contrast on $m = 0$ does not imply that those scattering volumes contain no faults; it follows from the phase-insensitive condition described above. Likewise, changes in the visibility of higher order $m = 0$ peaks are assigned to the growth-dependent mosaic and resolution state rather than to lateral-registry disorder.

The ordered baseline and the transition-model parameterization are summarized in Table 1. Numerical values from the superseded scalar two-population fit are not carried into the new model because they do not map uniquely onto three population-specific ϵ_j values and three normalized weights.

Table 1: PbI_2 ordered-baseline parameters and stacking-disorder parameterization for the selected-rod comparison. Numerical transition-matrix parameters must be reported from a refit using Eqs. (20)–(24); values from the superseded scalar model are not reused.

Parameter class	Fitted value or model quantity	Literature comparison	Role in the selected-rod model
Lattice and coordinates	$a = 4.5583$ Å, $c = 6.985$ Å; Pb at $(0, 0, 0)$; I1 at $(1/3, 2/3, 0.2677)$ and I2 at $(2/3, 1/3, 0.7323)$.	Lattice constants remain within the 2H single-crystal structural spread, and the iodine coordinate follows the corrected $z \simeq 0.268$ value.[42, 43]	Fixes the ordered Bragg positions and the $\lvert F_{hk}(L)\rvert^2$ envelope before stacking disorder is introduced.
Occupancies	A weakly floating test returned $o_{\mathrm{Pb}} = 0.999$, $o_{\mathrm{I1}} = 1.001$, and $o_{\mathrm{I2}} = 1.000$; the ordered baseline was then evaluated with full occupancies.	The clean ordered model uses full occupancies; the density-deficient occupancy reported for a Palosz-type sample is sample-specific and was not used as the default model.[43, 47]	Keeps the PbI_2 average structure stoichiometric while the diffuse rod intensity is assigned to stacking correlations.
Directional ADPs	$U_R/U_z = 0.0105/0.0185$ for Pb and $0.016/0.024$ Å^2 for I1 and I2.	Site-resolved Pb and I ADPs are not widely or consistently reported for direct transfer; the qualitative inequality $U_z(\mathrm{Pb}) > U_R(\mathrm{Pb})$ is retained.[42, 48]	Moderates relative ordered intensities and is interpreted as a sample-specific displacement refinement rather than a universal PbI_2 displacement constant.
Transition-matrix stacking disorder	ϵ_{2H}, ϵ_{4H}, ϵ_{6H}; normalized weights w_{2H}, w_{4H}, w_{6H}; and finite stack length N. Final numerical values are pending the transition-matrix refit.	These are effective selected-rod descriptors under the stated parent rules and equal-alternative fault templates, not universal PbI_2 constants.	Controls the redistribution of the ordered rod envelope into broadened maxima, finite-thickness fringes, and diffuse between-peak intensity.

The physical interpretation is therefore deliberately conditional on the model. Each ϵ_j measures the probability of departing from the selected parent event within population j, the w_j describe the incoherent population mixture, and N controls the finite-stack interference. Numerical values should be inserted only after the experimental profiles have been refit with this parameterization.

5 Refinement workflow

The workflow is placed after the ordered and PbI_2 result sections because the reader has now seen what the parameters are meant to explain. The refinement is staged rather than fully global from the beginning. Many parameters affect similar detector-space observables: detector distance shifts peak positions, sample alignment changes the apparent incident condition, mosaicity changes profile widths and off-condition intensity, and structure factors change relative peak heights. A simultaneous fit can therefore hide physical errors by compensating one parameter group with another. The staged workflow fixes the most instrumental quantities first, then introduces sample orientation, mosaicity, ordered intensity, and finally stacking disorder where needed.

The first distinction is between instrumental parameters and sample parameters. Instrumental parameters describe the beam and detector and should be shared across samples collected under the same experimental configuration. Sample parameters describe the film:

incidence condition, sample offset, mosaic distribution, structural intensity, and, for PbI_2, stacking disorder. The refinement sequence in Table 2 uses the detector-space observable most sensitive to each group before that group is used in the final image projection.

Table 2: Staged refinement workflow. Steps 1–5 are used for the ordered Bi_2Se_3 and Bi_2Te_3 validation. Step 6 is the added PbI_2 extension for stacking-fault diffuse scattering. All sample images are dark-subtracted before the listed observables are formed, and the final measured/calculated comparison applies the same detector-space projection to data and simulation.

Step	Stage	Data	Parameters refined	Primary information
1	Beam characterization	Direct-beam images at several detector distances	Beam widths w_x, w_z and divergences $\Delta\theta_x$, $\Delta\theta_z$	Incident phase space
2	Detector geometry calibration	Powder calibrant image	Detector distance D, detector tilts (β, γ), and beam center (x_0, y_0)	Detector mapping
3	Sample alignment and goniometer misalignment	Indexed reflection centers from sample images	Sample geometry $(\theta_i, \delta, z_B, z_S)$, in-plane rotation χ, goniometer misalignment (α, ψ)	Detector-space peak positions
4	Mosaicity/profile refinement	Selected local detector ROIs, specular-tail checks, and local profile shapes	Gaussian-plus-Lorentzian mosaic parameters	Center-aligned profile shapes, off-specular arcs, and specular tails
5	Ordered-film validation	Selected detector-space Bragg regions and Q_z projections from fixed-m trajectories	Image scale factors and ordered structure-factor parameters	Relative Bragg intensities and measured/calculated line-shape agreement
6	PbI_2 diffuse extension	Fixed-Q_R diffuse regions, ordered-baseline residuals, and projected Q_z profiles	Stacking-fault probabilities or layer-pair correlation parameters	Diffuse intensity along Q_z and separation of ordered and disorder-derived scattering

The instrumental stages fix where an ideal reflection should appear before any sample-specific physics is refined. Direct-beam images define the incident phase space, including beam width and angular divergence. A powder calibrant fixes the detector mapping: distance, beam center, and detector tilts. These two stages are intentionally separated from sample fitting because an incorrect detector geometry can otherwise be absorbed into apparent sample tilt, mosaic broadening, or shifted reciprocal-space trajectories.

The alignment stage uses indexed sample reflections to determine how the film sits in the beam. This stage fixes the nominal incident angle, sample offsets, in-plane rotation, and goniometer misalignment well enough that calculated Bragg-rod trajectories can be overlaid on the detector. Only after this step does it make sense to refine the mosaic distribution, because mosaicity should explain profile shape and off-condition intensity rather than compensate for a misplaced trajectory.

The mosaicity stage constrains the shared Gaussian-plus-Lorentzian angular distribution. Local off-specular profiles constrain the narrow core, while the specular-family tails and low-L features test the long-tail component. The ordered-film validation then applies the same projection to measured and calculated images and compares profiles along fixed-m trajectories. At that point the structure-factor parameters and image scale factors are constrained by relative intensities, not by detector placement alone.

For PbI_2, the ordered workflow is not discarded. It provides the geometry, projection, and mosaic baseline needed before diffuse scattering can be interpreted. The additional stacking-disorder stage asks whether a physically defined layer-correlation model accounts for the remaining diffuse intensity along fixed-Q_R cylinders. This keeps the PbI_2 result tied to the same detector-space framework while making clear which extra parameters belong specifically to stacking faults.

Implementation details, including lookup tables, Monte Carlo sampling, sub-pixel remapping, parameter-correlation plots, and the propagated uncertainty from 2θ–ϕ and Q transformations, should be documented in the supporting information. The main-text role of the workflow is to make the logic of the refinement reproducible: determine instrumental geometry first, determine sample orientation second, refine mosaic/profile effects third, and only then interpret ordered or diffuse intensity differences as structural information.

6 Discussion and conclusion

The ordered Bi_2Se_3 and Bi_2Te_3 films validate the mosaic model and detector-space forward model. A single calculation accounts for detector geometry, finite beam phase space, sample alignment, mosaicity, resolution, and ordered structure-factor weighting, then compares measured and calculated profiles along the same fixed-m trajectories. The key result is direct line-shape agreement in detector-derived projections, including features that would be difficult to interpret from a one-dimensional reduction alone.

Keeping the comparison in detector space is essential because the constraining intensity is distributed among off-specular arcs, cap-like $m = 0$ features, weak off-condition specular peaks, and bright low-L specular-region intensity that a conventional one-dimensional reduction would obscure. Because the calculated image includes the same effects and is projected in the same way as the measurement, the overlay tests the complete experimental model. The off-condition $m = 0$ peaks show why a narrow Gaussian-like mosaic core is insufficient and why the Lorentzian-like tail is required, while the near-origin low-L feature emphasizes the importance of finite beam and wavelength phase space.

The PbI_2 section outlines how the same framework can be extended from ordered Bragg line-shape validation to diffuse scattering from stacking disorder. In that extension, the detector geometry, beam phase space, mosaicity, and projection procedure remain the same, but the reciprocal-space intensity is generated from a stacking-fault correlation model rather than from a perfectly periodic ordered stack. This separation is important: diffuse intensity should be interpreted as stacking information only after geometry and mosaicity have been constrained by the ordered-film workflow.

The practical refinement workflow is therefore part of the scientific argument. Instrumental parameters are determined first, sample alignment second, mosaic and profile parameters third, and ordered or diffuse intensity parameters last. This ordering reduces parameter compensation and makes the final data/model comparison interpretable. The resulting manuscript structure follows the same logic: show the detector features that require each model ingredient, explain the physical effect in reciprocal-space terms, and then use measured/calculated overlays as the validation.

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